UNITED STATES PATENT APPLICATION

PROVISIONAL PATENT APPLICATION

Title of Invention:

ARTIFICIAL INTELLIGENCE-INTEGRATED VEHICLE CALIBRATION, DIAGNOSTIC, AND COMMUNICATION BUS ANALYSIS SYSTEM WITH CONTEXT-AWARE ADVISORY, CROSS-OFFSET CALIBRATION TRANSPLANTATION, AND MULTI-TIER REASONING-BASED FAULT DETECTION

Inventor(s): [INVENTOR FULL LEGAL NAME]

Assignee: PPEI, LLC (or applicable entity)

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CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Patent Application No. [TO BE ASSIGNED], filed [DATE], the entire disclosure of which is incorporated herein by reference.

ABSTRACT

A computer-implemented system and method for vehicle electronic control unit (ECU) calibration editing, diagnostic analysis, and communication bus monitoring, comprising: (a) a calibration editor that parses industry-standard definition files and binary firmware images, automatically aligns memory offsets between files of differing base addresses using a multi-strategy alignment engine, and enables calibration value transplantation between binaries with non-matching memory layouts; (b) an artificial intelligence advisory module that receives real-time context from the calibration editor including the currently selected calibration map, ECU platform identifier, alignment state, and user modification history, and generates context-specific engineering guidance regarding safe operating parameters, interdependent calibration relationships, and potential failure modes; © a diagnostic reasoning engine that analyzes time-series vehicle telemetry data, detects anomalies across correlated parameter channels, generates multi-step causal reasoning chains explaining probable root causes, and produces annotated fault-region visualizations overlaid on time-series graphs; (d) a tiered calibration visibility system that automatically classifies calibration maps into skill-based access tiers using pattern-matching against map identifiers, enabling the same binary and definition file to present different views to users of different expertise levels; and (e) an intelligent communication bus analyzer that captures raw Controller Area Network (CAN) frames in real time, cross-references captured arbitration identifiers against a multi-manufacturer module address database, detects byte-level data changes between successive frames, and applies artificial intelligence to suggest signal definitions for previously unidentified frames.

FIELD OF THE INVENTION

The present invention relates generally to vehicle electronic control systems, and more particularly to computer-implemented systems and methods for editing ECU calibration data, performing vehicle diagnostics, analyzing vehicle communication bus traffic, and providing artificial intelligence-assisted guidance for vehicle calibration and repair operations.

BACKGROUND OF THE INVENTION

The Problem with Existing Vehicle Diagnostic and Calibration Tools

Modern vehicles contain dozens of electronic control units (ECUs) that govern engine management, transmission control, body electronics, safety systems, and emissions compliance. Each ECU contains firmware consisting of an operating system and a calibration data segment. The calibration data segment contains thousands of individual parameters — lookup tables, scalar values, and multi-dimensional maps — that define the vehicle's operating behavior.

Modifying these calibration parameters is essential for vehicle performance optimization, aftermarket component integration (such as larger fuel tanks, different tire sizes, or upgraded turbocharger systems), emissions system service, and diagnostic troubleshooting. Currently, this work requires specialized desktop software applications such as EFILive, HP Tuners, WinOLS, INCA, FORScan, and AlphaOBD.

These existing tools suffer from several fundamental limitations:

First, existing calibration editors provide no intelligent guidance. When a user opens a calibration map containing 500 cells of numerical data, the tool displays the raw numbers with no explanation of what the map controls, what values are safe, how the map interacts with other maps, or what the consequences of a given change might be. The user must possess extensive prior knowledge of ECU calibration theory, or risk damaging the engine, transmission, or emissions system through uninformed modifications.

Second, existing tools cannot transplant calibration data between binary files that have different memory base addresses. When a vehicle manufacturer releases an

updated operating system, the calibration data segment may shift to a different memory region. A calibration developed for one operating system version cannot be directly applied to another version because the byte offsets do not match. Currently, this requires manual identification and remapping of every calibration parameter — a process that can take hours or days for a professional calibration engineer.

Third, existing diagnostic tools present fault codes (Diagnostic Trouble Codes, or DTCs) as isolated events without causal analysis. A vehicle may report DTC P0087 (Fuel Rail Pressure Too Low), but the tool does not analyze the correlated telemetry data to determine whether the root cause is a failing high-pressure fuel pump, a restricted fuel filter, a leaking pressure relief valve, or an incorrect calibration target. The user must perform this reasoning manually.

Fourth, existing calibration tools present all calibration parameters in a flat, unfiltered list. A typical ECU definition file contains 3,000 to 15,000 calibration maps. A vehicle owner who wants to change a speed limiter must navigate the same interface as a calibration engineer who needs access to injector compensation curves. No existing tool provides skill-based filtering that adapts the interface to the user's expertise level.

Fifth, existing CAN bus analysis tools (Vehicle Spy, SavvyCAN, CANoe) display raw hexadecimal frame data without intelligent interpretation. The user must manually identify which ECU module is broadcasting each frame, decode the data bytes using external signal databases, and identify anomalous patterns through visual inspection. No existing tool applies artificial intelligence to automatically identify modules, decode signals, and interpret bus traffic patterns.

The present invention addresses all five of these limitations through an integrated system that combines calibration editing, diagnostic analysis, communication bus monitoring, and artificial intelligence advisory capabilities.

SUMMARY OF THE INVENTION

The present invention provides a computer-implemented system and method for vehicle ECU calibration, diagnostics, and communication bus analysis, comprising five principal inventive aspects:

Inventive Aspect 1: Context-Aware AI Calibration Advisory System. An artificial intelligence module that operates within a calibration editing environment and

receives real-time contextual information including the ECU platform identifier, the currently loaded definition file metadata, the currently selected calibration map (including its name, description, data type, axis definitions, physical value range, and current values), the binary file alignment state, and the user's modification history. The AI module processes this contextual information together with a domain-specific knowledge base encompassing ECU architecture, calibration theory, tuning interdependencies, and failure modes, and generates context-specific engineering guidance in natural language. The guidance addresses safe operating parameter ranges for the specific ECU platform and map, interdependent relationships between the selected map and other calibration parameters, potential failure modes resulting from proposed modifications, and step-by-step calibration strategies for achieving specific performance objectives.

Inventive Aspect 2: Cross-Offset Calibration Segment Transplantation with Multi-Strategy Alignment. A method for transplanting calibration data between binary firmware images that have different memory base addresses, comprising: (a) parsing a definition file that maps calibration parameter names to memory addresses; (b) independently determining the memory offset for each binary file using a multi-strategy alignment engine that sequentially attempts format-embedded address extraction, known ECU family offset validation, and brute-force offset search with physical value validation scoring; © reading calibration values from a source binary using the source offset; and (d) writing the calibration values to a target binary using the target offset, such that the calibration data is correctly placed in the target binary's memory layout regardless of differences in base addresses between the source and target files.

Inventive Aspect 3: Multi-Step Causal Reasoning Diagnostic Engine with Annotated Fault Visualizations. A method for analyzing vehicle telemetry data comprising: (a) receiving time-series data containing multiple correlated parameter channels; (b) applying anomaly detection algorithms that evaluate each parameter against expected ranges and against correlated parameters; © generating multi-step causal reasoning chains that trace the logical sequence from observed anomaly to probable root cause, considering multiple competing hypotheses; and (d) producing time-series visualizations with color-coded fault region overlays and reasoning chain annotations, such that the user receives both a visual indication of where the fault occurred and a logical explanation of why it occurred.

Inventive Aspect 4: Tiered Calibration Map Visibility System. A method for automatically classifying calibration maps into skill-based access tiers comprising: (a)

receiving a set of calibration map definitions from a parsed definition file; (b) applying a hierarchical pattern-matching engine that evaluates each map's identifier against a plurality of pattern groups associated with functional categories and skill tiers; © assigning each map to one of a plurality of skill tiers ranging from basic (suitable for vehicle owners) to expert (suitable for calibration engineers); and (d) presenting a filtered view of calibration maps based on the selected skill tier, such that the same definition file and binary produce different user interfaces for users of different expertise levels.

Inventive Aspect 5: Intelligent Communication Bus Analyzer with AI-Assisted Frame Identification. A system for real-time vehicle communication bus analysis comprising: (a) a hardware bridge that captures raw CAN frames and transmits them to a browser-based application via WebSocket; (b) a frame processing engine that tracks per-arbitration-ID statistics including frame rate, byte-level delta detection, and directional classification; © a module identification engine that cross-references captured arbitration IDs against a multi-manufacturer database of known ECU module addresses; and (d) an artificial intelligence module that analyzes byte-change patterns in unidentified frames and suggests probable signal definitions based on observed data characteristics and known ECU behavior patterns.

DETAILED DESCRIPTION OF THE INVENTION

System Architecture Overview

Referring now to the drawings, FIG. 1 illustrates the overall system architecture of the present invention. The system comprises seven interconnected layers organized in a hierarchical stack:

Layer 1 — Hardware Bridge Layer (110). The hardware bridge layer provides physical connectivity to the vehicle's communication bus. In the preferred embodiment, a USB-connected CAN interface device (such as a PEAK PCAN-USB adapter) is connected to the vehicle's On-Board Diagnostics (OBD-II) port. A bridge application (112) running on the user's local computer communicates with the CAN interface device using the python-can library and exposes a WebSocket server (114) that the browser-based application connects to. The bridge application handles CAN frame transmission and

reception, OBD-II protocol encapsulation (including ISO 15765-2 transport protocol for multi-frame messages), and raw CAN frame monitoring.

Layer 2 — Protocol Engine Layer (120). The protocol engine layer implements the vehicle communication protocols including OBD-II (ISO 15031-5), Unified Diagnostic Services (UDS, ISO 14229), ISO-TP transport protocol (ISO 15765-2), and Controller Area Network (CAN 2.0A and 2.0B). This layer also implements manufacturer-specific security access algorithms (seed/key computation) for General Motors, Ford, Stellantis (RAM/Chrysler), BRP (CAN-am/Sea-Doo), Polaris, and Kawasaki platforms. The protocol engine translates high-level diagnostic requests (e.g., “read DID F190”) into properly formatted CAN frames and reassembles multi-frame responses.

Layer 3 — Live Data Layer (130). The live data layer provides real-time vehicle telemetry acquisition. This includes an OBD-II datalogger (132) that polls configurable sets of Parameter Identifiers (PIDs) at specified intervals, an IntelliSpy CAN bus monitor (134) that captures all raw CAN frames on the bus, and a drag strip timing module (136) that extracts acceleration event timing from telemetry data. The live data layer supports 400+ PIDs across automotive (GM, Ford, RAM) and powersports (CAN-am, Sea-Doo, Polaris, Kawasaki) platforms.

Layer 4 — Vehicle Services Layer (140). The vehicle services layer provides guided UDS-based service operations including module scanning (142), vehicle configuration coding (144), service procedures (146), and VIN modification (148). The module scanner queries all standard ECU addresses on the CAN bus and reads identification Data Identifiers (DIDs) from each responding module. Vehicle configuration coding reads and writes manufacturer-specific configuration blocks (such as Ford IPC as-built data) to modify parameters including fuel tank capacity, tire size correction factor, and body control module features. Service procedures execute multi-step UDS command sequences for operations including diesel particulate filter regeneration, fuel injector coding, tire pressure monitoring sensor relearn, and transmission adaptive value reset.

Layer 5 — Calibration Editor Layer (150). The calibration editor layer is described in detail in the following sections and constitutes a primary inventive aspect of the present invention.

Layer 6 — Analysis and Reasoning Layer (160). The analysis and reasoning layer processes telemetry data through a diagnostic engine (162) that applies anomaly detection, correlation analysis, and multi-step causal reasoning to produce fault reports with annotated visualizations. This layer also includes a health scoring module

(164) that grades the vehicle across eight functional systems on a quantitative scale, and a performance analysis module (166) that generates dynamometer-style power curves and drag strip timing data from telemetry.

Layer 7 — AI Intelligence Layer (170). The AI intelligence layer comprises the Erika advisory module (172), which is described in detail in the following sections. The AI module interfaces with all other layers, receiving contextual data from the calibration editor, diagnostic analyzer, communication bus monitor, and vehicle services modules, and providing natural language guidance to the user.

Detailed Description of Inventive Aspect 1: Context-Aware AI Calibration Advisory System

Referring to FIG. 2, the context-aware AI calibration advisory system (200) operates within the calibration editor environment and comprises three principal components: a context aggregation module (210), a domain knowledge base (220), and a language model interface (230).

Context Aggregation Module (210). The context aggregation module continuously monitors the state of the calibration editor and assembles a structured context object that is transmitted to the language model with each user query. The context object comprises:

(a) **ECU Platform Context (212):** The ECU family identifier (e.g., “GM_E41_L5P”), the processor type (e.g., “Infineon Tricore TC297”), the byte order (MSB_FIRST or MSB_LAST), the flash memory base address, and the calibration segment boundaries.

(b) **Definition File Context (214):** The source file name, the total number of calibration maps parsed, the total number of measurement channels, the parse time, and the detected ECU family.

© **Alignment Context (216):** The computed memory offset between the definition file addresses and the binary file addresses, the alignment confidence score (0.0 to 1.0), and the alignment method used (format-embedded, known-offset, or brute-force).

(d) **Selected Map Context (218):** The currently selected calibration map’s name, description, functional category, data type, physical unit, axis definitions (names, units, breakpoint values), current physical values, value range (minimum and maximum), and modification state.

(e) **Modification History Context (219):** The set of maps that have been modified in the current session, including the original and modified values for each changed cell.

The context aggregation module serializes this information into a structured prompt that is prepended to the user's natural language query before transmission to the language model.

Domain Knowledge Base (220). The domain knowledge base comprises a curated corpus of vehicle calibration engineering knowledge organized into the following categories:

(a) **ECU Architecture Knowledge (222):** Detailed specifications for each supported ECU platform including processor architecture, memory layout, communication protocols, security access levels, and known calibration segment boundaries.

(b) **Calibration Theory Knowledge (224):** Functional descriptions of calibration map categories including fuel injection control (rail pressure targets, injection timing, injector compensation), boost/turbocharger control (boost pressure targets, variable geometry turbocharger position maps, wastegate duty cycles), torque management (torque request limits, torque reduction strategies, traction control intervention), transmission control (shift schedules, torque converter clutch apply pressures, line pressure), and emissions systems (exhaust gas recirculation rates, diesel particulate filter regeneration thresholds, selective catalytic reduction dosing).

© **Tuning Interdependency Knowledge (226):** Documented relationships between calibration parameters, including the interdependence of boost pressure targets and fuel rail pressure targets, the relationship between injection timing advance and peak cylinder pressure, the interaction between torque management limits and transmission shift quality, and the effect of exhaust gas recirculation rate on combustion temperature and NOx production.

(d) **Failure Mode Knowledge (228):** Documented failure modes associated with calibration modifications, including turbocharger overspeeding from excessive boost targets, high-pressure fuel pump failure from excessive rail pressure commands, piston/ring damage from excessive injection timing advance, and transmission clutch failure from excessive torque multiplication.

(e) **CAN Bus Protocol Knowledge (229):** Comprehensive knowledge of vehicle communication protocols, diagnostic services, CAN bus analysis tool concepts

(including terminology and workflows from industry-standard tools), and manufacturer-specific module addressing schemes.

Language Model Interface (230). The language model interface transmits the assembled context and user query to a large language model and returns the generated response to the user interface. The interface is configured to instruct the language model to provide responses that are specific to the ECU platform and calibration map identified in the context, that reference specific numerical ranges appropriate for the identified platform, that identify interdependent calibration parameters by name, and that warn of specific failure modes associated with proposed modifications.

Novel Operation. The combination of real-time editor state injection (context aggregation) with a domain-specific calibration knowledge base and a language model interface produces an advisory system that generates guidance specific to the exact calibration parameter the user is currently editing, on the exact ECU platform they are working with, considering the exact modifications they have already made. This is distinguished from generic AI chatbots that lack calibration context, and from existing calibration tools that lack any AI advisory capability.

Detailed Description of Inventive Aspect 2: Cross-Offset Calibration Segment Transplantation

Referring to FIG. 3, the cross-offset calibration segment transplantation system (300) enables the transfer of calibration data between binary firmware images that have different memory base addresses. The system comprises a definition parser (310), a multi-strategy alignment engine (320), and a transplantation executor (330).

Definition Parser (310). The definition parser reads industry-standard ASAM MCD-2MC (A2L) definition files or equivalent calibration definition formats (such as Cummins CSV exports) and produces a structured representation of the calibration data layout. For each calibration map, the parser extracts: the map name and description, the memory address as specified in the definition file, the data type and byte size, the record layout (axis structure, data direction), the computation method for converting between raw binary values and physical engineering values, and the axis point definitions (shared, fixed, or inline).

The parser supports the following computation methods: rational function (RAT_FUNC), interpolation table (TAB_INTP), identical pass-through (IDENTICAL),

linear scaling (LINEAR), and formula-based (FORM). The parser also supports the following axis types: common axis (COM_AXIS) with shared axis point references, fixed axis (FIX_AXIS) with computed breakpoints, and standard axis (STD_AXIS) with inline breakpoints.

Multi-Strategy Alignment Engine (320). The alignment engine determines the memory offset between the addresses specified in the definition file and the actual byte positions in a given binary file. The engine employs three strategies in sequential order, proceeding to the next strategy only if the current strategy fails to produce a high-confidence result:

Strategy 1 — Format-Embedded Address Extraction (322). Binary files in Motorola S-Record format (.ptp, .srec, .s19, .s28, .s37) and Intel HEX format (.hex, .ihex) contain address information embedded in each data record. The alignment engine extracts the base address from the file format and computes the offset as the difference between the definition file's reference address space and the format-embedded base address. This strategy produces a confidence score of 0.85 to 0.95 when the file format is recognized.

Strategy 2 — Known ECU Family Offset Validation (324). For recognized ECU families, the alignment engine maintains a database of common flash memory base addresses (e.g., 0x94400000 for GM E41, 0x80000000 for Bosch MG1C, 0x00060000 for GM T93). For each candidate base address, the engine computes the corresponding offset, reads a sample set of VALUE-type calibration maps from the binary using that offset, converts the raw values to physical values using the definition file's computation methods, and scores the candidate based on how many sample values fall within the definition file's specified minimum and maximum physical value ranges. The candidate with the highest score is selected. This strategy produces a confidence score of 0.70 to 0.90 when a valid candidate is found.

Strategy 3 — Brute-Force Offset Search with Physical Value Validation (326). When neither of the above strategies produces a high-confidence result, the engine performs a systematic search across offset candidates at configurable alignment boundaries (default: 0x1000 byte boundaries). For each candidate offset, the engine reads sample calibration maps and scores the candidate using the same physical value validation method as Strategy 2. The search space is bounded by the binary file size and the definition file's address range. The highest-scoring candidate is selected. This strategy produces a confidence score of 0.40 to 0.80 depending on the number of validating samples.

Transplantation Executor (330). Once the alignment engine has determined independent offsets for a source binary (332) and a target binary (334), the transplantation executor performs the following steps for each calibration map to be transplanted:

(a) Compute the source byte position as: `source_position = map_address - source_offset`

(b) Read the raw byte values from the source binary at the source byte position, using the data type and byte count specified in the definition file.

© Compute the target byte position as: `target_position = map_address - target_offset`

(d) Write the raw byte values to the target binary at the target byte position.

(e) If the source and target binaries have different byte orders (as determined by the ECU family or definition file metadata), perform byte-order conversion before writing.

Novel Operation. The use of a definition file as a platform-independent reference frame, combined with independent offset determination for each binary file, enables calibration data transplantation between files whose memory layouts differ due to operating system version changes, manufacturer updates, or format conversion. This is distinguished from byte-level copy operations that require identical memory layouts, and from manual offset remapping that requires expert knowledge of each binary's memory structure.

Detailed Description of Inventive Aspect 3: Multi-Step Causal Reasoning Diagnostic Engine

Referring to FIG. 4, the diagnostic reasoning engine (400) analyzes vehicle telemetry data and produces annotated fault visualizations with causal reasoning chains. The engine comprises a telemetry ingestion module (410), an anomaly detection module (420), a reasoning chain generator (430), and a visualization annotator (440).

Telemetry Ingestion Module (410). The ingestion module accepts time-series telemetry data in comma-separated value (CSV) format, as produced by common vehicle datalogging tools (HP Tuners, EFILive) or by the system's own datalogger. The module parses column headers to identify parameter channels, normalizes

timestamps to a common time base, and resamples channels to a uniform sample rate when necessary.

Anomaly Detection Module (420). The anomaly detection module evaluates each parameter channel against multiple criteria:

(a) **Absolute Range Violations (422):** Each parameter is compared against predefined acceptable ranges. For example, fuel rail pressure below 200 bar during commanded injection is flagged as anomalous.

(b) **Rate-of-Change Violations (424):** Sudden changes in parameter values that exceed physiologically plausible rates are flagged. For example, coolant temperature increasing by more than 5 degrees Celsius per second indicates a sensor fault or catastrophic cooling system failure.

© **Cross-Channel Correlation Violations (426):** Related parameters are evaluated for expected correlations. For example, commanded fuel rail pressure and actual fuel rail pressure should track within a defined tolerance band. Divergence beyond the tolerance indicates a fuel system fault. Similarly, commanded boost pressure and actual boost pressure should correlate; divergence indicates a boost leak, turbocharger fault, or wastegate malfunction.

(d) **Pattern-Based Detection (428):** The module applies pattern recognition to identify known fault signatures. For example, oscillating exhaust gas temperature (EGT) spread between cylinders with a specific frequency pattern indicates a failing fuel injector in the cylinder with the highest EGT deviation.

Reasoning Chain Generator (430). For each detected anomaly, the reasoning chain generator produces a multi-step causal explanation comprising:

(a) **Observation Statement (432):** A description of the detected anomaly, including the parameter name, the anomalous value or pattern, and the timestamp range.

(b) **Hypothesis Generation (434):** A set of competing hypotheses that could explain the observed anomaly, ranked by probability based on the specific parameter values, correlated channel data, and known failure mode frequencies.

© **Evidence Evaluation (436):** For each hypothesis, the generator evaluates supporting and contradicting evidence from other parameter channels. For example, if the hypothesis is “fuel filter restriction” for a low rail pressure anomaly, the generator checks whether fuel rail pressure recovers at lower fuel command rates (supporting

evidence) or remains low regardless of command rate (contradicting evidence favoring pump failure).

(d) **Conclusion Statement (438):** A ranked list of probable root causes with confidence levels and recommended diagnostic or repair actions.

Visualization Annotator (440). The visualization annotator produces time-series graphs with the following overlays:

(a) **Fault Region Highlighting (442):** Semi-transparent colored overlays on the time axis indicating the duration and severity of each detected anomaly. Color coding indicates fault category (red for critical, amber for warning, blue for informational).

(b) **Reasoning Chain Annotations (444):** Callout annotations positioned at the fault region boundaries that display the reasoning chain summary, including the observation, top hypothesis, and recommended action.

© **Cross-Channel Reference Lines (446):** Visual indicators connecting correlated parameters that are involved in the same fault detection, enabling the user to see the relationship between, for example, commanded and actual rail pressure during a fuel system fault.

Novel Operation. The combination of multi-criteria anomaly detection with multi-step causal reasoning and annotated fault visualizations produces a diagnostic output that explains not only what anomaly was detected and when it occurred, but why it occurred and what should be done about it. This is distinguished from existing diagnostic tools that display fault codes without causal analysis, and from existing datalog viewers that display time-series data without anomaly detection or reasoning.

Detailed Description of Inventive Aspect 4: Tiered Calibration Map Visibility System

Referring to FIG. 5, the tiered calibration map visibility system (500) automatically classifies calibration maps into skill-based access tiers and presents filtered views based on the selected tier. The system comprises a pattern matching engine (510), a tier assignment module (520), and a filtered presentation module (530).

Pattern Matching Engine (510). The pattern matching engine evaluates each calibration map's identifier (name string) against a hierarchical set of pattern groups. Each pattern group comprises one or more regular expression patterns associated

with a functional category and a skill tier. The pattern groups are organized by specificity, with more specific patterns evaluated before more general patterns.

In the preferred embodiment, the pattern groups include:

Tier	Functional Category	Example Patterns
1 (Basic)	Speed Limiters	`.*[Ss]peed.*[Ll]imit.*`, `.*[Vv]max.*`
1 (Basic)	Idle Speed	`.*[Ii]dle.*[Rr]pm.*`, `.*[Ii]dle.*[Ss]peed.*`
1 (Basic)	Rev Limiter	`.*[Rr]ev.*[Ll]imit.*`, `.*[Rr]pm.*[Ll]imit.*`
2 (Street)	Torque Management	`.*[Tt]orque.*[Ll]im.*`, `.*[Tt]q.*[Mm]ax.*`
2 (Street)	Boost Targets	`.*[Bb]oost.*[Tt]arget.*`, `.*[Bb]st.*[Dd]es.*`
2 (Street)	Rail Pressure Targets	`.*[Rr]ail.*[Pp]ress.*[Tt]arget.*`
3 (Advanced)	EGR Control	`.*[Ee][Gg][Rr].*`, `.*[Ee]xhaust.*[Rr]ecirc.*`
3 (Advanced)	VGT Control	`.*[Vv][Gg][Tt].*`, `.*[Tt]urbo.*[Cc]ontrol.*`
4 (Expert)	DPF/SCR Systems	`.*[Dd][Pp][Ff].*`, `.*[Ss][Cc][Rr].*`
4 (Expert)	OBD Monitors	`.*[Oo][Bb][Dd].*[Mm]onitor.*`
5 (Full)	All Remaining	Default assignment for unmatched maps

Tier Assignment Module (520). The tier assignment module processes each calibration map through the pattern matching engine and assigns the map to the lowest-numbered (most accessible) tier whose patterns match the map's identifier. Maps that do not match any specific pattern are assigned to Tier 5 (Full Access). The assignment is performed once when the definition file is parsed and cached for the duration of the editing session.

Filtered Presentation Module (530). The filtered presentation module maintains a user-selectable tier level and displays only those calibration maps whose assigned tier is less than or equal to the selected tier. The module updates the map tree browser, search results, and map count indicators to reflect the filtered set. When the user selects Tier 1 (Basic), only maps assigned to Tier 1 are visible. When the user selects

Tier 3 (Advanced), maps assigned to Tiers 1, 2, and 3 are visible. When the user selects Tier 5 (Full), all maps are visible.

Novel Operation. The automatic classification of calibration maps into skill-based tiers using pattern matching against map identifiers, combined with a filtered presentation that adapts the interface complexity to the user's selected expertise level, enables a single tool to serve users ranging from vehicle owners (who need access to 10-20 maps) to calibration engineers (who need access to 3,000+ maps). This is distinguished from existing calibration tools that present all maps in an unfiltered list regardless of user expertise.

Detailed Description of Inventive Aspect 5: Intelligent Communication Bus Analyzer

Referring to FIG. 6, the intelligent communication bus analyzer (600) captures, processes, and interprets raw CAN bus traffic in real time. The system comprises a frame capture module (610), a frame processing engine (620), a module identification engine (630), and an AI interpretation module (640).

Frame Capture Module (610). The frame capture module communicates with the hardware bridge layer to initiate and terminate bus monitoring sessions. Upon receiving a start command, the hardware bridge configures the CAN interface to receive all frames on the bus (promiscuous mode) and transmits each received frame to the browser application via WebSocket. Each frame message contains: the arbitration identifier (11-bit or 29-bit), the data length code (0-8 bytes), the raw data bytes, and a high-resolution timestamp.

Frame Processing Engine (620). The frame processing engine maintains a per-arbitration-ID state table that tracks:

(a) **Frame Statistics (622):** Total frame count, instantaneous frame rate (Hz), first-seen timestamp, and last-seen timestamp for each arbitration ID.

(b) **Byte-Level Delta Detection (624):** For each arbitration ID, the engine compares the current frame's data bytes against the previous frame's data bytes and identifies which byte positions have changed. Changed bytes are flagged for visual highlighting in the user interface.

© **Directional Classification (626):** Each arbitration ID is classified as a diagnostic request (0x700-0x7FF range, even addresses), a diagnostic response (0x700-0x7FF

range, odd addresses), or a broadcast/network management frame (all other ranges).

Module Identification Engine (630). The module identification engine maintains a multi-manufacturer database of known ECU module addresses. For each captured arbitration ID, the engine queries the database to determine if the ID corresponds to a known module. The database includes entries for multiple manufacturers:

(a) **Ford modules (632):** 48 modules including PCM, TCM, ABS, BCM, IPC, APIM, PSCM, RCM, PAM, IPMA, and others, with their standard CAN request and response arbitration IDs.

(b) **RAM/Stellantis modules (634):** 21 modules including PCM, TCM, ABS, BCM, IPC, TIPM, RFH, HVAC, and others.

© **GM modules (636):** 11 modules including ECM, TCM, EBCM, BCM, IPC, HPCM, and others.

When a match is found, the module name and manufacturer are associated with the captured frame for display in the user interface.

AI Interpretation Module (640). The AI interpretation module applies artificial intelligence to analyze captured bus traffic and provide interpretive guidance. The module operates in two modes:

(a) **Known Frame Interpretation (642):** For frames from identified modules, the AI module cross-references the data bytes against the UDS DID reference database to suggest which diagnostic data the frame may contain.

(b) **Unknown Frame Analysis (644):** For frames from unidentified arbitration IDs, the AI module analyzes byte-change patterns over time (which bytes change frequently, which are static, which correlate with vehicle operating conditions) and suggests probable signal definitions based on common CAN signal encoding patterns (e.g., a 16-bit big-endian value in bytes 0-1 that increases linearly with engine speed is likely an RPM signal).

Novel Operation. The combination of real-time CAN frame capture with automatic module identification, byte-level delta detection, and AI-assisted signal interpretation produces a bus analysis tool that automatically identifies which ECU modules are communicating, highlights data changes as they occur, and suggests signal definitions for unknown frames. This is distinguished from existing CAN bus analyzers that display raw hexadecimal data without automatic identification or intelligent interpretation.

CLAIMS

Independent Claims

Claim 1. A computer-implemented method for providing context-aware calibration guidance for a vehicle electronic control unit (ECU), the method comprising:

(a) parsing a calibration definition file to extract a plurality of calibration map definitions, each calibration map definition comprising a map name, a memory address, a data type, and a computation method for converting between raw binary values and physical engineering values;

(b) loading a binary firmware image and determining a memory offset between the definition file address space and the binary file address space;

© monitoring a calibration editor user interface to detect a user selection of a calibration map from the plurality of calibration map definitions;

(d) in response to detecting the user selection, assembling a context object comprising: an ECU platform identifier, the selected calibration map's name, description, data type, physical unit, axis definitions, current physical values, and modification state;

(e) transmitting the context object together with a user's natural language query to an artificial intelligence language model that has been configured with a domain-specific knowledge base comprising ECU architecture specifications, calibration theory, tuning interdependency relationships, and failure mode documentation; and

(f) receiving from the language model and presenting to the user a natural language response that is specific to the selected calibration map on the identified ECU platform, comprising at least one of: safe operating parameter ranges, interdependent calibration parameter relationships, potential failure modes, or calibration modification strategies.

Claim 2. A computer-implemented method for transplanting calibration data between binary firmware images having different memory base addresses, the method comprising:

(a) parsing a calibration definition file to extract a plurality of calibration map definitions, each comprising a map name and a memory address in a definition address space;

(b) receiving a source binary firmware image and determining a source memory offset between the definition address space and the source binary address space using a multi-strategy alignment engine, the multi-strategy alignment engine sequentially attempting: (i) extraction of a base address embedded in the source binary file format, (ii) validation of known ECU family base addresses by reading sample calibration maps and scoring physical value validity, and (iii) brute-force search across offset candidates with physical value validation scoring;

© receiving a target binary firmware image and independently determining a target memory offset between the definition address space and the target binary address space using the same multi-strategy alignment engine;

(d) for each calibration map to be transplanted: reading raw byte values from the source binary at a source position computed as the map's definition address adjusted by the source offset, and writing the raw byte values to the target binary at a target position computed as the map's definition address adjusted by the target offset;

whereby calibration data is correctly transplanted between binary files whose memory layouts differ due to operating system version changes, manufacturer updates, or format differences.

Claim 3. A computer-implemented method for analyzing vehicle telemetry data and generating annotated fault visualizations with causal reasoning chains, the method comprising:

(a) receiving time-series telemetry data comprising a plurality of correlated parameter channels recorded from a vehicle;

(b) applying anomaly detection to each parameter channel, the anomaly detection comprising at least two of: absolute range violation detection, rate-of-change violation detection, cross-channel correlation violation detection, and pattern-based fault signature detection;

© for each detected anomaly, generating a multi-step causal reasoning chain comprising: an observation statement describing the anomaly, a plurality of competing hypotheses ranked by probability, evidence evaluation for each hypothesis

based on correlated parameter channel data, and a conclusion statement identifying probable root causes with confidence levels;

(d) generating a time-series visualization comprising: the parameter channel data plotted against time, color-coded fault region overlays indicating the temporal extent and severity of each detected anomaly, and reasoning chain annotations positioned at fault region boundaries displaying the causal reasoning summary.

Claim 4. A computer-implemented method for presenting calibration maps at different levels of complexity based on user expertise, the method comprising:

(a) parsing a calibration definition file to extract a plurality of calibration map definitions, each comprising at least a map name identifier;

(b) applying a hierarchical pattern-matching engine to each map name identifier, the pattern-matching engine comprising a plurality of pattern groups, each pattern group associated with a functional category and a skill tier from a plurality of ordered skill tiers;

© assigning each calibration map to the most accessible skill tier whose associated pattern group matches the map's name identifier, and assigning unmatched maps to the highest skill tier;

(d) receiving a user-selected skill tier level;

(e) presenting in a user interface only those calibration maps whose assigned skill tier is less than or equal to the user-selected skill tier level;

whereby the same calibration definition file produces different interface views for users of different expertise levels.

Claim 5. A system for intelligent real-time vehicle communication bus analysis, the system comprising:

(a) a hardware bridge module configured to capture raw Controller Area Network (CAN) frames from a vehicle communication bus and transmit the frames to a browser-based application via a WebSocket connection;

(b) a frame processing engine configured to maintain per-arbitration-identifier statistics including frame count, frame rate, and byte-level delta detection between successive frames sharing the same arbitration identifier;

© a module identification engine configured to cross-reference captured arbitration identifiers against a multi-manufacturer database of known ECU module addresses and associate identified module names with captured frames;

(d) an artificial intelligence interpretation module configured to analyze byte-change patterns in frames from unidentified arbitration identifiers and generate suggested signal definitions based on observed data characteristics.

Dependent Claims

Claim 6. The method of Claim 1, wherein the domain-specific knowledge base further comprises knowledge of vehicle communication bus protocols, CAN bus analysis tool concepts, and manufacturer-specific module addressing schemes, enabling the AI language model to provide guidance regarding communication bus traffic interpretation in addition to calibration guidance.

Claim 7. The method of Claim 1, wherein the context object further comprises a modification history indicating which calibration maps have been modified in the current editing session and the magnitude of each modification, enabling the AI language model to consider cumulative modification effects when generating guidance.

Claim 8. The method of Claim 2, wherein the multi-strategy alignment engine further comprises a confidence scoring module that assigns a numerical confidence score to each alignment result, and the method further comprises presenting the confidence score to the user and issuing a warning when the confidence score falls below a predetermined threshold.

Claim 9. The method of Claim 2, wherein the calibration definition file is in ASAM MCD-2MC (A2L) format and the method further comprises supporting computation methods including rational function (RAT_FUNC), interpolation table (TAB_INTP), identical pass-through (IDENTICAL), and linear scaling (LINEAR) for converting between raw binary values and physical engineering values during transplantation validation.

Claim 10. The method of Claim 2, wherein the source binary and target binary are in different file formats selected from the group consisting of: raw binary, Motorola S-Record, Intel HEX, and proprietary container formats, and the method further comprises converting each file to a normalized byte array with a resolved base address before performing offset determination.

Claim 11. The method of Claim 3, wherein the cross-channel correlation violation detection comprises evaluating the relationship between a commanded parameter value and an actual parameter value, and detecting divergence beyond a tolerance band as indicative of a system fault in the actuator or sensor chain associated with the parameter.

Claim 12. The method of Claim 3, wherein the pattern-based fault signature detection comprises detecting oscillating temperature spread patterns across multiple exhaust gas temperature sensors and identifying the cylinder associated with the highest temperature deviation as a probable failing fuel injector location.

Claim 13. The method of Claim 3, further comprising generating a quantitative health score for each of a plurality of vehicle functional systems based on the number, severity, and duration of detected anomalies within each system, and presenting the health scores in a summary report.

Claim 14. The method of Claim 4, wherein the plurality of ordered skill tiers comprises at least: a basic tier suitable for vehicle owners, a performance tier suitable for enthusiast tuners, an advanced tier suitable for professional tuners, an expert tier suitable for calibration engineers, and a full-access tier providing unfiltered access to all calibration maps.

Claim 15. The method of Claim 4, wherein the pattern-matching engine recognizes manufacturer-specific naming conventions including General Motors prefix patterns, Bosch prefix patterns, and Cummins prefix patterns, enabling automatic tier assignment across multiple ECU manufacturer platforms.

Claim 16. The system of Claim 5, wherein the byte-level delta detection comprises comparing each byte of a current frame against the corresponding byte of the most recently received frame sharing the same arbitration identifier, and flagging changed bytes for visual highlighting in the user interface.

Claim 17. The system of Claim 5, wherein the multi-manufacturer database comprises module addresses for at least three vehicle manufacturers, and the module identification engine determines the vehicle manufacturer based on the set of responding arbitration identifiers observed during a monitoring session.

Claim 18. The system of Claim 5, further comprising a session recording module configured to store all captured frames with timestamps to a persistent storage medium, and a session replay module configured to replay stored sessions with the

same frame processing, module identification, and AI interpretation applied to the replayed frames.

Claim 19. An integrated vehicle intelligence system comprising the method of Claim 1, the method of Claim 2, the method of Claim 3, the method of Claim 4, and the system of Claim 5, wherein the AI language model of Claim 1 further receives diagnostic reasoning chains generated by the method of Claim 3 and communication bus analysis results generated by the system of Claim 5 as additional context for generating calibration guidance.

Claim 20. The integrated system of Claim 19, further comprising a vehicle service procedure engine configured to execute guided multi-step Unified Diagnostic Services (UDS) command sequences through the hardware bridge module, the service procedures including at least: diesel particulate filter forced regeneration via RoutineControl service, fuel injector coding via WriteDataByIdentifier service, and tire pressure monitoring sensor relearn via manufacturer-specific UDS sequences.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating the seven-layer system architecture of the vehicle intelligence platform, showing the interconnections between the hardware bridge layer, protocol engine layer, live data layer, vehicle services layer, calibration editor layer, analysis and reasoning layer, and AI intelligence layer.

FIG. 2 is a block diagram illustrating the context-aware AI calibration advisory system, showing the context aggregation module, domain knowledge base, and language model interface, with data flow arrows indicating how editor state is assembled into a context object and transmitted to the language model.

FIG. 3 is a flowchart illustrating the cross-offset calibration segment transplantation process, showing the definition parser, the multi-strategy alignment engine with its three sequential strategies, and the transplantation executor that reads from the source binary and writes to the target binary using independently determined offsets.

FIG. 4 is a block diagram illustrating the diagnostic reasoning engine, showing the telemetry ingestion module, anomaly detection module (with four detection criteria), reasoning chain generator (with four reasoning steps), and visualization annotator (with three overlay types).

FIG. 5 is a diagram illustrating the tiered calibration map visibility system, showing how a single definition file containing 3,000+ maps is filtered through the pattern matching engine and tier assignment module to produce different views for five skill tiers, with example map counts at each tier.

FIG. 6 is a block diagram illustrating the intelligent communication bus analyzer, showing the frame capture module, frame processing engine, module identification engine with multi-manufacturer database, and AI interpretation module with known-frame and unknown-frame analysis modes.

FIG. 7 is a data flow diagram illustrating how all system modules interconnect, showing the closed-loop data flow from datalogger through diagnostic analyzer through AI advisor through calibration editor through binary export through ECU and back to datalogger.

NOTES FOR PATENT ATTORNEY

Prior Art Considerations

The following existing products and publications should be considered during prior art search:

1. **ETAS INCA** — Professional calibration tool with A2L support and measurement capabilities. Does not include AI advisory, tiered visibility, or cross-offset transplantation.

2. **Vector CANape** — Professional measurement and calibration tool. Does not include AI advisory or diagnostic reasoning.

3. **EFILive** — Aftermarket calibration tool for GM vehicles. Does not include A2L parsing, AI advisory, cross-offset transplantation, or diagnostic reasoning.

4. **HP Tuners VCM Suite** — Aftermarket calibration and diagnostic tool. Does not include A2L parsing, AI advisory, cross-offset transplantation, or tiered visibility.

5. **WinOLS** — ECU calibration tool with hex editor and map finder. Does not include AI advisory, diagnostic reasoning, or communication bus analysis.

6. **FORScan** — Ford diagnostic and configuration tool. Does not include calibration editing, AI advisory, or diagnostic reasoning.
7. **Intrepid Vehicle Spy** — CAN bus analysis tool. Does not include AI-assisted frame identification or calibration editing.
8. **PEAK SavvyCAN** — Open-source CAN bus analyzer. Does not include AI interpretation or module identification.

Recommended Filing Strategy

1. **File a provisional patent application immediately** to establish priority date. The provisional should cover all five inventive aspects.
2. **Consider filing the non-provisional as a single application with all five aspects** (strongest position — the integrated system claim in Claim 19 is the most defensible because the combination is clearly novel even if individual aspects face prior art challenges).
3. **Alternatively, consider filing as a continuation-in-part** with the integrated system as the parent and individual aspects as divisional applications, if the attorney determines that any individual aspect is strong enough to stand alone.
4. **The seed/key algorithms should NOT be included in the patent** — these are better protected as trade secrets since patent disclosure would reveal the computation methods to competitors.
5. **The specific calibration map pattern groups (Tier assignment patterns) may benefit from trade secret protection** rather than patent disclosure, as they represent accumulated domain expertise that competitors would need to independently develop.

Inventor Declaration

The undersigned inventor(s) declare that they are the original and first inventor(s) of the subject matter claimed herein, and that the information provided in this application is true and correct to the best of their knowledge.

Signature: _____

Printed Name: _____

Date: _____

This document is a draft prepared for attorney review. It is not a filed patent application. All claims, descriptions, and drawings descriptions should be reviewed and refined by a registered patent attorney before filing with the United States Patent and Trademark Office.